

ORIGINAL ARTICLE

Regression of Left Ventricular Hypertrophy in Primary Aldosteronism

Jessica Goi¹, Jacopo Burrello², Holger Schneider³, Erik Johann Petersenn⁴, Stephanie Zopp⁵, Virginia Capuzzo, Ana María García Cano⁶, Laura Caja Guayerbas⁷, Elena Mena Ribas⁸, Alicia Sanmartín Sánchez⁹, Paola Parra Ramírez¹⁰, Patricia Martín Rojas-Marcos¹¹, Miguel Paja Fano¹², Patricia Díaz Guardiola, Ángel Rebollo Román¹³, Eva María Moya Mateo¹⁴, Jorge Gabriel Ruiz-Sanchez¹⁵, Marga González-Boillos¹⁶, Fernando Jaén Aguila¹⁷, Carolina María Perdomo¹⁸, Cristina Robles Lázaro, Mònica Recasens, Rebeca Barahona San Millán¹⁹, Emilia Gómez-Hoyos²⁰, María José Picón César²¹, Sonsoles Gutiérrez-Medina²², Cristina Lamas²³, Anna Casteras, Paolo Mulatero²⁴, Martin Reincke²⁵, Marta Araujo-Castro²⁶; on behalf of the Adrenal Working Group of the Spanish Society of Endocrinology & Nutrition (SEEN)

BACKGROUND: Left ventricular hypertrophy (LVH) is frequently identified in patients with primary aldosteronism; determinants of LVH regression after targeted treatment are debated. We assessed LVH prevalence before and after adrenalectomy or mineralocorticoid receptor antagonist (MRA) therapy, identified determinants of LVH regression, and evaluated the relationship between patients' outcomes and cardiac remodeling.

METHODS: We retrospectively analyzed an international cohort of patients with primary aldosteronism from 36 referral centers in Spain, Italy, and Germany. Echocardiographic recordings were included at baseline and after at least 1 year of targeted treatment. Regression models were applied to identify parameters associated with LVH regression and changes in left ventricular mass index.

RESULTS: Among 1454 patients (mean age, 51 years; 58.7% male) with primary aldosteronism, baseline LVH prevalence was 49.4%. Aldosterone, blood pressure, male sex, number of antihypertensive medications, and diabetes were associated with LVH. Of 473 patients with a follow-up echocardiography, 33.6% underwent adrenalectomy, 61.7% received MRA, and 4.7% were taking MRA postsurgery. After a median follow-up of 51 months, LVH regression was observed in 54.2% of patients (61.9% after adrenalectomy versus 48.6% with MRA treatment; $P=0.043$), and left ventricular mass index significantly decreased after both surgery and medical therapy. Higher pretreatment blood pressure and aldosterone levels were associated with a lower likelihood of LVH regression, regardless of treatment modality. Complete biochemical success was associated with greater left ventricular mass index reduction at follow-up.

CONCLUSIONS: LVH regressed in over half the patients after targeted treatment, with left ventricular mass index decreasing after both adrenalectomy and MRA therapy. Baseline aldosterone levels were associated with LVH regression, independently of blood pressure load. (*Hypertension*. 2026;83:00–00. DOI: 10.1161/HYPERTENSIONAHA.126.26909.) • **Supplement Material.**

Key Words: adrenalectomy ■ aldosterone ■ blood pressure ■ middle aged ■ mineralocorticoid receptor antagonists

Primary aldosteronism (PA) is the main cause of secondary hypertension, with a prevalence of ≈5% to 10% in unselected hypertensive populations^{1,2} and an even higher prevalence among patients with

resistant³ or severe⁴ hypertension. Beyond blood pressure (BP) elevation, PA is associated with an increased risk of cardiovascular morbidity and mortality compared with essential hypertension.^{5–7}

Correspondence to: Marta Araujo-Castro, Department of Endocrinology and Nutrition, Hospital Universitario Ramón y Cajal, Colmenar Viejo St km 9. PC, 28034 Madrid, Spain. Email marta.araujo@salud.madrid.org

Supplemental Material is available at <https://www.ahajournals.org/doi/suppl/10.1161/HYPERTENSIONAHA.126.26909>.

© 2026 American Heart Association, Inc.

Hypertension is available at www.ahajournals.org/journal/hyp

NOVELTY AND RELEVANCE

What Is New?

This study is based on the largest cohort of patients focused on the evaluation of left ventricular hypertrophy (LVH) in primary aldosteronism (PA).

We systematically applied PA surgical outcomes and PA medical treatment outcomes criteria and evaluated, for the first time, their association with LVH regression after adrenalectomy or mineralocorticoid receptor antagonist therapy.

What Is Relevant?

Higher baseline aldosterone levels were associated with LVH identification at baseline and predicted a lower likelihood of posttreatment regression of LVH, independently of blood pressure load.

Targeted treatment of PA promoted LVH regression in more than half of the patients.

Left ventricular mass significantly decreased after both surgical and medical treatments.

Patient stratification according to clinical and biochemical responses defined by PA surgical outcomes and PA medical treatment outcomes criteria may help clinicians identify patients who are more likely to achieve LVH regression.

Clinical/Pathophysiological Implications?

In patients with PA, baseline and follow-up echocardiographic evaluations should be routinely performed. Targeted treatment of PA, including adrenalectomy and mineralocorticoid receptor antagonist therapy, promotes significant reductions in left ventricular mass index and can lead to regression of LVH.

Nonstandard Abbreviations and Acronyms

AVS	adrenal venous sampling
BP	blood pressure
DDD	defined daily dose
LV	left ventricular
LVH	left ventricular hypertrophy
LVMI	left ventricular mass index
MRA	mineralocorticoid receptor antagonist
PA	primary aldosteronism
PAC	plasma aldosterone concentration
PAMO	primary aldosteronism medical treatment outcomes
PASO	primary aldosteronism surgical outcomes

Left ventricular hypertrophy (LVH) represents a cardinal manifestation of hypertension-mediated organ damage and is associated with adverse cardiovascular outcomes.^{8–10} Chronic aldosterone excess in PA promotes myocardial hypertrophy and fibrosis through mechanisms that extend beyond BP-related volume overload alone,^{11–13} resulting in greater left ventricular (LV) mass and a higher prevalence of LVH compared with essential hypertension.^{14,15} The enhanced cardiac remodeling contributes substantially to the elevated cardiovascular risk observed in this population.⁶

The management of PA relies on 2 therapeutic approaches: unilateral adrenalectomy for patients with lateralized aldosterone overproduction, and lifelong mineralocorticoid receptor antagonist (MRA) therapy for those with a bilateral phenotype or who are not eligible

for surgical intervention.¹⁶ Adrenalectomy improves BP control and leads to full biochemical cure in most cases.¹⁷ Several observational studies have reported a significant reduction in LV mass index (LVMI) after surgery, underscoring the potential for substantial reversal of adverse cardiac remodeling.^{14,18–22} MRAs can effectively reduce BP and normalize renin and potassium levels¹⁶ and have also been associated with a decrease in LVMI over time,^{14,20,21,23,24} particularly after at least 1 year of therapy.¹⁴ However, other studies have not consistently observed a significant improvement in cardiac structure after MRA therapy,^{18,19} and 3 recent meta-analyses suggest that the magnitude of LVMI reduction achieved with medical treatment may be less pronounced than that observed following surgery.^{25–27} Comparative analyses of parameters associated with LV mass normalization after adrenalectomy versus MRA therapy are scarce and limited by small sample sizes, single-center design, or relatively short follow-up.^{18,19,23,28,29} To address these gaps, we conducted an international cohort study involving 1454 patients with confirmed PA, managed across tertiary hypertension centers in Spain, Germany, and Italy. The aims of this study were: (1) to determine the prevalence of LVH at the diagnosis of PA and identify baseline parameters associated with its identification; (2) to assess the impact of standardized clinical and biochemical outcomes—as defined by PA surgical outcomes (PASO) and PA medical treatment outcomes (PAMO) criteria^{17,30}—on the echocardiographic evolution of LV structure after surgical or medical targeted treatment; and (3) to evaluate determinants of LVH regression after at least 1 year of PA-specific treatment.

METHODS

A complete description of data collection and patient outcome definitions is available in the [Supplemental Methods](#) section of the [Online-Only Data Supplement](#). All data and supporting materials are available within the article and its [Online Supplemental Files](#).

Study Design and Study Population

We retrospectively analyzed patients diagnosed with PA between January 1990 and June 2025 in 3 European countries: Germany (Munich), Italy (Turin), and Spain (34 centers belonging to the SPAIN-ALDO Registry³¹). Men and women aged 18 to 90 years with a confirmed diagnosis of PA and available baseline echocardiographic evaluation were eligible for the cross-sectional analysis. For the longitudinal analysis, patients were additionally required to have a follow-up echocardiography performed after at least 1 year of targeted treatment (adrenalectomy or MRA therapy). Patients without baseline echocardiographic assessment, without follow-up echocardiographic data for the longitudinal analysis, or who did not receive targeted treatment were excluded ([Figure S1](#)).

The study was approved by the local Research Ethics Committees and adhered to the principles of the Declaration of Helsinki. Patient consent was waived due to the retrospective and noninterventional nature of the study.

Diagnosis of PA and Subtype Differentiation

In accordance with the 2016 Endocrine Society recommendations,³² the diagnosis of PA was based on screening with the aldosterone-to-renin ratio followed by confirmation through at least 1 suppression test (seated saline infusion test, captopril-challenge test, oral sodium loading test or fludrocortisone suppression test), except for those cases presenting with overt PA (plasma aldosterone concentration [PAC] >20 ng/dL with K⁺ <3.5 mmol/L and plasma renin activity <0.20 ng/mL per hour [or direct renin concentration <5.0 mU/L]).³³

Differentiation between unilateral and bilateral disease was achieved using adrenal venous sampling (AVS), with results available in 66.6% (n=969/1454) of the study population (80.5% of patients with both baseline and follow-up data). The remaining patients had unavailable results or did not undergo AVS. Subtype diagnosis was defined by center-specific thresholds.

Targeted Treatment

Of the 473 patients with follow-up data, 61.7% (n=292) were treated with MRA therapy, 33.6% (n=159) underwent adrenalectomy, and the remaining patients (n=22) received MRA after surgery due to suboptimal postoperative outcome (the latter were excluded from comparisons between medical and surgical treatment). Twenty-six patients with unilateral disease at AVS subsequently declined surgery, while 18 patients underwent adrenalectomy according to imaging findings despite bilateral or undetermined PA subtype at AVS.

Clinical and biochemical responses to targeted treatment were classified as complete, partial, or absent according to PASO¹⁷ and PAMO³⁰ criteria for surgically treated patients and those receiving MRA therapy, respectively.

Definition of LVH

The echocardiographic parameters evaluated were LVMI, defined as LV mass adjusted for body surface area, LV ejection fraction, interventricular septum, and LV posterior wall diastolic dimension. LVH was defined as LVMI >95 g/m² in women and >115 g/m² in men, according to the recommendations of the American Society of Echocardiography and the European Association of Cardiovascular Imaging.³⁴ LVH was categorized as mild (LVMI 96–108 g/m² in women and 116–131 g/m² in men), moderate (LVMI 109–121 g/m² in women and 132–148 g/m² in men), or severe (LVMI >121 g/m² for women and >148 g/m² for men).³⁴ Regression of LVH after treatment was defined as normalization of LVMI, whereas those who did not achieve LVMI normalization after surgical or medical treatment were categorized as nonregressors.

Statistical Analysis

IBM SPSS Statistics 29 (IBM, New York), GraphPad Prism 9.0 (La Jolla, CA), and R (R Core Team, Vienna, Austria), with the RStudio integrated development environment (Posit Software), were used for statistical analyses and figure drafting. In cross-sectional analyses, nonnormally distributed variables were reported as the median and interquartile range and compared by the Mann-Whitney *U* test or the Kruskal-Wallis test, whereas normally distributed variables were expressed as mean±SD and analyzed by the Student *t* test, or 1-way ANOVA, as appropriate. Matched longitudinal analyses of quantitative variables were performed using the Wilcoxon signed-rank test or paired *t* test. Categorical variables were reported as absolute numbers and percentages and compared using the χ^2 or Fisher exact test. Univariable and stepwise multivariable linear or logistic regression models were performed to assess the association between clinical/biochemical parameters and percentage change in LVMI after treatment, and to identify factors independently associated with baseline LVH or with its regression after at least 1 year of targeted treatment. For multivariable models, all parameters associated with the outcome at univariable analysis were entered in subsequent steps (excluding collinear ones). In addition, a predefined set of clinically relevant variables was included a priori, as potential confounders. As a sensitivity analysis to account for potential selection bias related to the availability of follow-up echocardiography, we performed a missing-at-random inverse-probability-of-observation weighting. First, a logistic regression model was fitted in the entire baseline cohort (n=1454) to estimate the probability of having an available follow-up echocardiography, using baseline covariates available for both patients with and without follow-up data (including age, sex, referral center, presence of LVH, systolic BP, number of antihypertensive drugs, aldosterone and potassium levels, and known cardiovascular disease). Inverse-probability weights derived from the predicted probabilities were then applied to the longitudinal cohort (n=473), and the multivariable models assessing parameters associated with LVH regression and percentage change in LVMI were repeated using weighted logistic and linear regression models, respectively.

A *P* < 0.05 was considered significant. For multiple comparisons, *P* values were adjusted using the Bonferroni post hoc correction.

RESULTS

Prevalence of LVH at Baseline

A total of 1454 patients with baseline echocardiographic evaluation were included in the study: 516 from the SPAIN-ALDO Registry (Spain), 516 from the Munich center (Germany), and 422 from the Turin center (Italy; [Figure S1](#)). Baseline characteristics of the study population are reported in Table 1. The mean age was 51 years, 58.7% were men, and among patients with available subtype information, 47.1% ($n=457/969$) had unilateral PA; the average LVMI was 107 g/m². At baseline, 49.4% of patients ($n=719/1454$) had LVH, most commonly of mild (47.5%) or moderate (30.4%) severity.

Compared with patients without cardiac hypertrophy, those with LVH were older and more frequently men, had higher body mass index, BP, creatinine, glycated hemoglobin A1c, fasting plasma glucose, baseline and postseated saline infusion test PAC, lower estimated glomerular filtration rate and potassium levels, longer duration of hypertension, and a greater prevalence of hypertension grade II–III, hypokalemia, nodule(s) at imaging, unilateral PA (along with higher lateralization indices at AVS without cosyntropin stimulation), smoking habit, diabetes, dyslipidemia, obstructive sleep apnea, chronic kidney disease, and known cardiovascular disease ($P<0.01$ for all comparisons; Table 1). The prevalence of concomitant mild autonomous cortisol secretion did not differ between patients with and without LVH.

As expected, patients with LVH had higher LVMI (133 versus 86 g/m²), as well as greater interventricular septum (13 versus 11 mm) and LV posterior wall thickness (12 versus 10 mm), compared with those without cardiac remodeling ($P<0.001$ for all comparisons; Table 1).

At univariable logistic regression, clinical and biochemical parameters that differed between patients with and without LVH were associated with its presence at baseline. In the final multivariable model, the baseline factors that remained independently associated with the presence of LVH were male sex, higher systolic BP, greater antihypertensive treatment requirement, presence of diabetes, and higher PAC (Table S1 and Figure 1A).

Patient Characteristics After Stratification for Targeted Treatment

Of the initial cohort, 473 patients had an available echocardiographic evaluation after at least 1 year of targeted treatment ([Figure S1](#)). Differences in clinical, biochemical, radiological, and echocardiographic baseline and follow-up parameters between surgically ($n=159$) and medically treated patients ($n=292$) are reported in [Tables S2](#) and [S3](#). Among patients receiving medical therapy, spironolactone and canrenone were the most commonly used MRAs (81.5%), and the median defined daily dose (DDD) of MRA therapy was 1.33.

At baseline, patients who underwent adrenalectomy had a higher prevalence of LVH (61% versus 48.6%) and a consistently higher LVMI (113 versus 104 g/m²).

Overall, the median duration of follow-up was 51 months (38 months in the adrenalectomy group and 58 months in the MRA group; $P=0.010$). At follow-up, adrenalectomized patients displayed a higher rate of complete clinical (20.1% versus 14.4%) and biochemical outcomes (76.2% versus 47.6%) as compared with medically treated subjects; the medical treatment group tended to exhibit a higher prevalence of LVH (34.9% versus 27%; $P=0.087$) and a lower LVMI reduction ($P=0.073$).

Longitudinal Comparison After Targeted Treatment

Clinical, biochemical and echocardiographic parameters before and after treatment are shown in Table 2. In the overall cohort ($n=473$), we observed a reduction in LVMI, accompanied by a marked decrease in both the prevalence of LVH (53.1% versus 31.9%; $P<0.001$) and its severity ([Figure 1B](#) through [1D](#)). At follow-up, a higher proportion of patients exhibited a mild LVH phenotype (42.2% versus 56.8%), with a lower rate of severe LVH (26.4% versus 17%; $P=0.020$). A significant reduction was also observed in systolic/diastolic BP, estimated glomerular filtration rate, interventricular septum thickness, LV posterior wall dimension, and the median number of antihypertensive medications, while renin, potassium, and creatinine levels increased (Table 2; $P<0.01$ for all comparisons). Similar results were found after stratification by type of targeted treatment (adrenalectomy or MRA therapy; Table 2 and [Figure 2A](#) through [2C](#)); however, no significant reductions in LV wall thickness were detected among medically treated patients.

In patients who underwent surgical intervention ([Table S4](#) and [Figure 2D](#) and [2E](#)), complete clinical responders had a lower prevalence of LVH and a minor septum and LV posterior wall diastolic dimensions at baseline, as compared with partial or absent responders. Patients with a partial clinical outcome showed greater variation in interventricular septum thickness after surgery compared with absent responders, while no differences in the decrease in LVH prevalence or LVMI were found among the clinical outcome groups. After stratification for PASO biochemical outcomes, echocardiographic parameters did not differ at baseline or follow-up, although there was a trend toward lower rates of LVH in complete responders at both timeframes.

Among patients who received MRA therapy ([Table S5](#) and [Figure 2D](#) and [2E](#)), complete clinical responders tended to exhibit a lower prevalence of LVH at baseline and showed a significantly lower rate of LVH after treatment (10% versus 35.3%–41.4%; $P=0.002$). At

Table 1. Baseline Characteristics of the Study Population

Variable	Overall cohort (N=1454)	Absence of LVH (n=735)	Presence of LVH (n=719)	P value
Clinical parameters				
Age at referral, y	51±11.1	49±11.1	53±10.8	<0.001*
Female sex, n (%)	600 (41.3)	349 (47.5)	251 (34.9)	<0.001*
BMI, kg/m ²	28.1±5.25	27.4±5.33	28.8±5.07	<0.001*
BSA, m ²	1.97±0.254	1.94±0.262	2.00±0.241	<0.001*
SBP, mm Hg	152±21.4	149±19.7	156±22.5	<0.001*
DBP, mm Hg	93±12.7	92±12.0	94±13.4	0.008*
Duration of HTN, y	7 (2–15)	5 (1–12)	10 (3–16)	<0.001*
HTN grade II or III, n (%)	688 (52.2)	280 (42.6)	408 (61.7)	<0.001*
Antihypertensive drugs, n	2 (1–3)	2 (1–3)	3 (2–4)	<0.001*
Hypokalemia at any time, n (%)	841 (57.8)	380 (51.7)	461 (64.1)	<0.001*
Potassium supplementation, n (%)	385 (26.9)	165 (22.8)	220 (31.3)	<0.001*
Current smoking, n (%)	202 (16.9)	78 (13.3)	124 (20.3)	0.001*
Obesity, n (%)	461 (32.1)	201 (27.5)	260 (36.9)	<0.001*
Diabetes type 1 or 2, n (%)	224 (15.4)	83 (11.8)	141 (20.3)	<0.001*
Dyslipidemia, n (%)	540 (38.1)	233 (32.4)	307 (44.1)	<0.001*
Obstructive sleep apnea, n (%)	257 (19.1)	99 (14.5)	158 (23.8)	<0.001*
Chronic kidney disease, n (%)	89 (6.1)	24 (3.3)	65 (9.0)	<0.001*
Known CV disease, n (%) [*]	345 (23.7)	152 (20.7)	193 (26.8)	0.006*
Concurrent MACS, n (%)	108 (14.4)	56 (14.4)	52 (14.5)	0.949
Biochemical parameters				
Fasting plasma glucose, mg/dL	103±23.2	100±20.3	106±25.6	<0.001*
HbA1c, %	5.6±0.81	5.5±0.71	5.7±0.89	<0.001*
Creatinine, mg/dL	0.92±0.376	0.87±0.371	0.96±0.377	<0.001*
eGFR, mL/min	93±19.2	97±16.9	90±20.7	<0.001*
Total cholesterol, mg/dL	190±40.0	191±38.5	189±41.5	0.341
LDL cholesterol, mg/dL	115±34.9	116±34.0	114±35.8	0.253
HDL cholesterol, mg/dL	54±15.8	57±16.1	52±15.1	<0.001*
Triglycerides, mg/dL	117±72.5	109±70.5	125±73.8	<0.001*
Aldosterone, ng/dL	24.5 (16.8–37.4)	22.1 (15.4–34.4)	27.4 (18.5–40.1)	<0.001*
PRA, ng/mL per h	0.20 (0.17–0.40)	0.20 (0.10–0.40)	0.20 (0.20–0.40)	0.061
DRC, mU/L	2.8 (2.0–6.1)	2.7 (2.0–6.0)	2.9 (2.0–6.2)	0.803
ARR (PRA)	123 (69–221)	125 (69–212)	121 (67–227)	0.669
ARR (DRC)	6.3 (3.4–14.4)	6.0 (3.1–11.6)	6.9 (3.6–17.4)	0.060
Potassium, mmol/L	3.7±0.58	3.7±0.56	3.6±0.59	<0.001*
Lowest potassium, mmol/L	3.4±0.63	3.5±0.62	3.3±0.63	<0.001*
Suppression tests				
Suppression test data available, n (%)	1190 (81.8)	...	...	...
Type of suppression test, n (%)				
Seated SIT	926 (77.8)	497 (80.8)	429 (74.6)	0.024*
CCT	249 (20.9)	113 (18.4)	136 (23.7)	
Others [‡]	15 (1.3)	5 (0.8)	10 (1.7)	
Aldosterone after SIT, ng/dL	11.6 (8.1–19.5)	10.6 (7.7–16.4)	13.3 (8.9–23.0)	<0.001*
Aldosterone after CCT, ng/dL	18.5 (12.5–28.2)	16.9 (12.2–29.0)	19.2 (12.8–27.6)	0.385
PRA after SIT, ng/mL per h	0.15 (0.10–0.24)	0.15 (0.10–0.22)	0.15 (0.10–0.30)	0.473
PRA after CCT, ng/mL per h	0.20 (0.15–0.32)	0.20 (0.15–0.33)	0.20 (0.15–0.30)	0.528
DRC after SIT, mU/L	2.0 (2.0–4.1)	2.0 (2.0–3.9)	2.0 (2.0–4.4)	0.503
DRC after CCT, mU/L	2.6 (1.4–5.7)	2.6 (1.3–4.3)	2.6 (1.4–6.4)	0.542

(Continued)

Table 1. Continued

Variable	Overall cohort (N=1454)	Absence of LVH (n=735)	Presence of LVH (n=719)	P value
Imaging (CT scan and MRI)				
Presence of nodules, n (%)	839 (61.0)	377 (54.7)	462 (67.3)	<0.001*
Largest size of nodules, mm	15 (10–20)	15 (10–20)	14 (10–20)	0.880
Adrenal venous sampling				
PA subtype based on AVS, n (%)				
Unilateral	457 (47.1)	211 (42.5)	246 (52.2)	<0.001*
Bilateral	362 (37.4)	215 (43.2)	147 (31.1)	
Undetermined	150 (15.5)	71 (14.3)	79 (16.7)	
LI without ACTH	4.2 (1.8–13.6)	3.4 (1.8–12.0)	5.0 (2.1–14.5)	0.016*
LI with ACTH	5.0 (1.9–13.6)	5.1 (1.6–15.0)	5.0 (2.1–12.1)	0.849
Echocardiographic parameters				
Presence of LVH, n (%)	719 (49.4)	...	...	...
Severity of LVH, n (%)				
Mild	...	...	301 (47.5)	...
Moderate	...	...	193 (30.4)	...
Severe	...	...	140 (22.1)	...
LV mass index, g/m ²	107±33.4	86±16.0	133±29.4	<0.001*
IVS thickness, mm	12±2.4	11±1.9	13±2.2	<0.001*
PW thickness, mm	11±2.2	10±1.7	12±2.1	<0.001*
LV ejection fraction, %	63±8.1	63±7.8	63±8.4	0.156

Baseline patient characteristics in the overall cohort and stratified by presence of LVH on echocardiography. ACTH indicates adrenocorticotropic hormone; ARR, aldosterone-to-renin ratio; AVS, adrenal venous sampling; BMI, body mass index; CCT, captopril-challenge test; CT, computed tomography; CV, cardiovascular; DBP, diastolic blood pressure; DRC, direct renin concentration; eGFR, estimated glomerular filtration rate; HbA1c, glycated hemoglobin A1c; HDL, high density lipoprotein; HTN, hypertension; IVS, interventricular septum; LDL, low density lipoprotein; LI, lateralization index; LV, left ventricular; LVH, left ventricular hypertrophy; MACS, mild autonomous cortisol secretion; MRI, magnetic resonance imaging; PA, primary aldosteronism; PRA, plasma renin activity; PW, posterior wall; SBP, systolic blood pressure; and SIT, saline infusion test.

*Excluding left ventricular hypertrophy.

†Oral sodium loading test or fludrocortisone suppression test.



baseline, they also had lower LVMI and reduced inter-ventricular septal thickness values compared with partial or absent responders, while no differences were observed in the improvement of echocardiographic parameters. No significant differences were found after stratification for PAMO biochemical outcomes, although complete responders tended to have a lower prevalence of LVH at follow-up as compared with partial or absent responders.

Regression of LVH After Targeted Treatment

The overall rate of LVH regression was 54.2% (n=136/251), with a higher proportion of patients achieving regression in the adrenalectomy group than in the medical treatment cohort (61.9% versus 48.6%; Table S3 and Figure 3A). No differences were found after stratification for surgical and medical outcomes, either across clinical or biochemical responses (Tables S4 and S5; Figure 3B and 3C).

At baseline, compared with patients in the nonregression group (n=115/251), those who achieved LVMI normalization had lower systolic BP, less severe

hypertension, a lower prevalence of dyslipidemia, lower PAC, and exhibited a milder baseline LVH phenotype with lower echocardiographic values (Table 3). At follow-up, patients with LVH regression had lower BP, required fewer antihypertensive drugs, and displayed a greater reduction in the antihypertensive treatment requirement compared with those with persistent LVH (Table S6).

At univariable logistic regression analysis, LVH regression was associated with several baseline factors, including lower systolic BP, less severe hypertension, lower prevalence of dyslipidemia, lower PAC, lower LVMI, and a milder LVH phenotype, as well as with a complete clinical response to treatment. In the multivariable model, baseline systolic BP and PAC independently predicted LVH regression after adjustment for baseline LVH severity, type of treatment, follow-up duration, and treatment responses (Table S7 and Figure 3D). No association was observed between MRA DDD and LVH regression after medical treatment (odds ratio, 0.91 [95% CI, 0.61–1.36]; $P=0.656$).

We then evaluated the parameters associated with LVMI changes after surgical or medical treatment (Table S8 and Figure 3E). At univariable linear regression

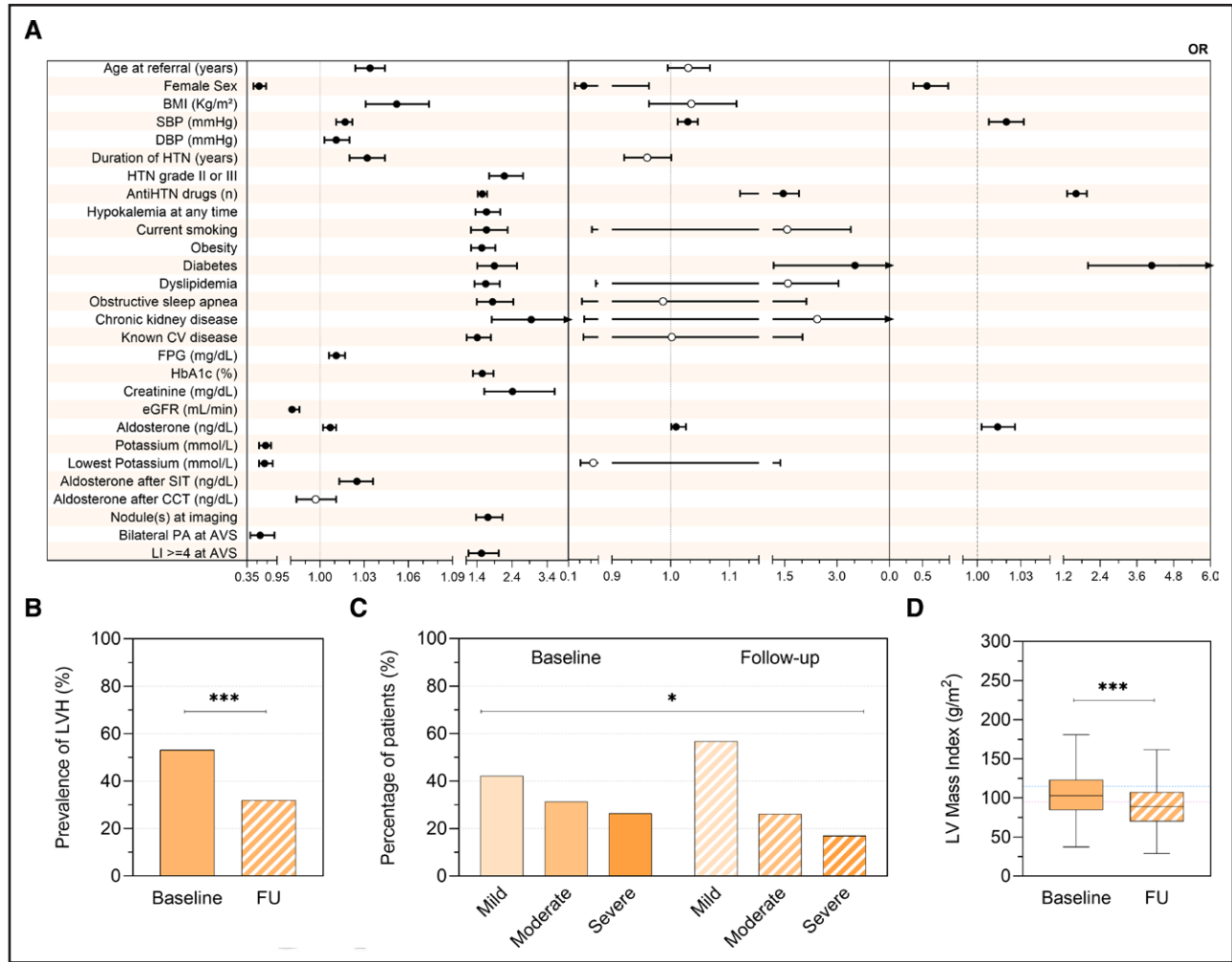


Figure 1. Prevalence of left ventricular hypertrophy (LVH) and associated clinical and biochemical parameters. **A**, Factors associated with LVH at baseline evaluation. The panel shows odds ratios (ORs) with 95% CIs from univariable and stepwise multivariable logistic regression analyses (first box from the left; univariable; second and third boxes: Step 1 and Step 2 of the adjusted models, respectively; see also Table S1). **B**, Prevalence of LVH in the overall cohort at baseline and follow-up evaluation. **C**, LVH severity. **D**, Left ventricular (LV) mass index (g/m²). **P*<0.05; ****P*<0.001. AVS indicates adrenal venous sampling; BMI, body mass index; CCT, captopril-challenge test; CV, cardiovascular; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; FU, follow-up; HbA1c, glycated hemoglobin A1c; HTN, hypertension; LI, lateralization index; PA, primary aldosteronism; SBP, systolic blood pressure; and SIT, saline infusion test.

analysis, complete biochemical success, baseline LVH severity, treatment duration, and follow-up systolic BP, antihypertensive treatment requirement, renin, and potassium levels were associated with the percentage change in LVMI. After correction for confounding parameters, in the final multivariable model, only a lower follow-up systolic BP, a lower antihypertensive treatment requirement at follow-up, and a milder baseline LVH phenotype remained independently associated with a greater reduction in LVMI. Of note, when we restricted this analysis to patients receiving MRA therapy (Table S9), a higher MRA DDD was associated with a greater reduction in LVMI, independently of all the other parameters included in the model.

To evaluate whether the exclusion of patients without follow-up echocardiography may have biased the

analyses of the longitudinal cohort, we performed an inverse-probability weighting sensitivity analysis (Table S10). A logistic model was fitted in the full baseline cohort to estimate each patient's probability of undergoing follow-up echocardiography, including baseline clinical, biochemical, echocardiographic, and center-level covariates (not shown). Inverse-probability weights were then applied to the longitudinal cohort, and the multivariable models estimating parameters associated with LVH regression and the percentage change in LVMI were refitted using weighted regression models. The results of the weighted models were directionally consistent with the primary complete-case analyses, suggesting that differential availability of follow-up echocardiography did not significantly influence our findings, although residual selection bias related to unmeasured factors cannot be

Table 2. Longitudinal Changes in patients' Parameters With Available Echocardiography at Follow-Up ≥ 1 Year

Variable	Overall cohort (N=473)			Adrenalectomy (n=159)			MRA therapy (n=292)		
	Baseline	Follow-up	P value	Baseline	Follow-up	P value	Baseline	Follow-up	P value
Clinical parameters									
BMI, kg/m ²	27.7 $\pm$ 4.80	27.8 $\pm$ 4.94	0.099	27.8 $\pm$ 4.85	27.9 $\pm$ 4.88	0.728	27.4 $\pm$ 4.7	27.6 $\pm$ 4.91	0.065
Obesity, n (%)	132 (28.0)	133 (28.2)	0.920	45 (28.3)	43 (27.0)	0.806	78 (26.8)	79 (27.1)	0.920
SBP, mm Hg	152 $\pm$ 20.7	134 $\pm$ 15.9	<0.001*	154 $\pm$ 21.4	134 $\pm$ 14.8	<0.001*	150 $\pm$ 19.7	134 $\pm$ 16.1	<0.001*
DBP, mm Hg	93 $\pm$ 12.3	85 $\pm$ 10.6	<0.001*	94 $\pm$ 11.8	86 $\pm$ 10.2	<0.001*	93 $\pm$ 12.4	85 $\pm$ 10.6	<0.001*
Antihypertensive drugs, n	2 (1–3)	2 (1–3)	<0.001*	2 (1–3)	1 (0–2)	<0.001*	2 (1–3)	2 (1–3)	<0.001*
Biochemical parameters									
Creatinine, mg/dL	0.92 $\pm$ 0.345	1.06 $\pm$ 0.544	<0.001*	0.92 $\pm$ 0.399	1.11 $\pm$ 0.750	<0.001*	0.92 $\pm$ 0.322	1.04 $\pm$ 0.409	<0.001*
eGFR, mL/min	93 $\pm$ 19.1	79 $\pm$ 20.0	<0.001*	94 $\pm$ 20.1	79 $\pm$ 20.8	<0.001*	92 $\pm$ 18.7	80 $\pm$ 20.0	<0.001*
Fasting plasma glucose, mg/dL	102 $\pm$ 22.6	105 $\pm$ 22.9	0.065	102 $\pm$ 22.4	103 $\pm$ 24.1	0.651	102 $\pm$ 22.7	105 $\pm$ 21.7	0.051
HbA1c, %	5.5 $\pm$ 0.70	5.8 $\pm$ 0.80	<0.001*	5.5 $\pm$ 0.62	5.7 $\pm$ 0.75	<0.001*	5.5 $\pm$ 0.76	5.7 $\pm$ 0.84	<0.001*
Total cholesterol, mg/dL	194 $\pm$ 37.7	190 $\pm$ 42.3	0.120	192 $\pm$ 37.7	191 $\pm$ 40.1	0.790	195 $\pm$ 37.7	190 $\pm$ 43.9	0.177
LDL cholesterol, mg/dL	117 $\pm$ 33.9	114 $\pm$ 36.3	0.095	115 $\pm$ 33.5	114 $\pm$ 37.5	0.500	118 $\pm$ 34.2	115 $\pm$ 35.8	0.231
HDL cholesterol, mg/dL	56 $\pm$ 15.8	53 $\pm$ 14.9	<0.001*	56 $\pm$ 16.1	54 $\pm$ 17.4	0.179	56 $\pm$ 15.4	53 $\pm$ 13.1	<0.001*
Triglycerides, mg/dL	115 $\pm$ 66.0	132 $\pm$ 93.7	<0.001*	113 $\pm$ 69.4	140 $\pm$ 129.8	0.002*	116 $\pm$ 65.3	129 $\pm$ 69.0	<0.001*
Aldosterone, ng/dL	22.0 (14.1–34.1)	18.6 (10.0–32.8)	0.700	24.7 (17.3–42.9)	8.8 (6.0–12.7)	<0.001*	20.1 (13.3–30.2)	27.3 (19.8–45.5)	<0.001*
PRA, ng/mL per h	0.20 (0.20–0.40)	2.34 (0.45–4.38)	<0.001*	0.30 (0.20–0.60)	2.65 (1.55–5.13)	<0.001*	0.20 (0.17–0.40)	2.40 (0.40–5.08)	<0.001*
DRC, mU/L	3.3 (2.0–6.7)	19.5 (7.9–51.9)	<0.001*	2.7 (2.0–5.6)	18.0 (7.7–53.5)	<0.001*	3.5 (2.0–7.4)	22.2 (7.1–56.9)	<0.001*
ARR (PRA)	...	...	...	169 (74–264)	5 (2–7)	<0.001*	...	...	...
ARR (DRC)	...	...	...	7.9 (3.3–13.1)	0.4 (0.2–1.5)	<0.001*	...	...	...
Potassium, mmol/L	3.6 $\pm$ 0.55	4.4 $\pm$ 0.46	<0.001*	3.4 $\pm$ 0.55	4.5 $\pm$ 0.45	<0.001*	3.8 $\pm$ 0.50	4.4 $\pm$ 0.46	<0.001*
Echocardiographic parameters									
Presence of LVH, n (%)	251 (53.1)	151 (31.9)	<0.001*	97 (61.0)	43 (27.0)	<0.001*	142 (48.6)	102 (34.9)	<0.001*
Mild, n (%)	93 (42.2)	80 (56.8)	0.020*	34 (38.2)	22 (51.1)	0.073	54 (44.7)	57 (61.9)	0.040*
Moderate, n (%)	69 (31.4)	37 (26.2)		26 (29.2)	15 (34.9)		39 (32.2)	19 (20.7)	
Severe, n (%)	58 (26.4)	24 (17.0)		29 (32.6)	6 (14.0)		28 (23.1)	16 (17.4)	
LV mass index, g/m ²	107 $\pm$ 33.4	92 $\pm$ 28.9	<0.001*	113 $\pm$ 35.1	92 $\pm$ 27.0	<0.001*	104 $\pm$ 32.7	91 $\pm$ 29.0	<0.001*
IVS thickness, mm	12 $\pm$ 2.4	11 $\pm$ 2.5	0.001*	12 $\pm$ 2.2	11 $\pm$ 2.2	<0.001*	12 $\pm$ 2.5	11 $\pm$ 2.6	0.157
PW thickness, mm	11 $\pm$ 2.3	10 $\pm$ 4.7	0.001*	11 $\pm$ 1.9	11 $\pm$ 7.3	<0.001*	10 $\pm$ 2.5	10 $\pm$ 2.1	0.147

Matched longitudinal analysis in the overall cohort, and after stratification by type of targeted treatment (adrenalectomy or MRA therapy). ARR indicates aldosterone-to-renin ratio; BMI, body mass index; DBP, diastolic blood pressure; DRC, direct renin concentration; eGFR, estimated glomerular filtration rate; IVS, interventricular septum; LV, left ventricular; LVH, left ventricular hypertrophy; MRA, mineralocorticoid receptor antagonist; PRA, plasma renin activity; PW, posterior wall; and SBP, systolic blood pressure.

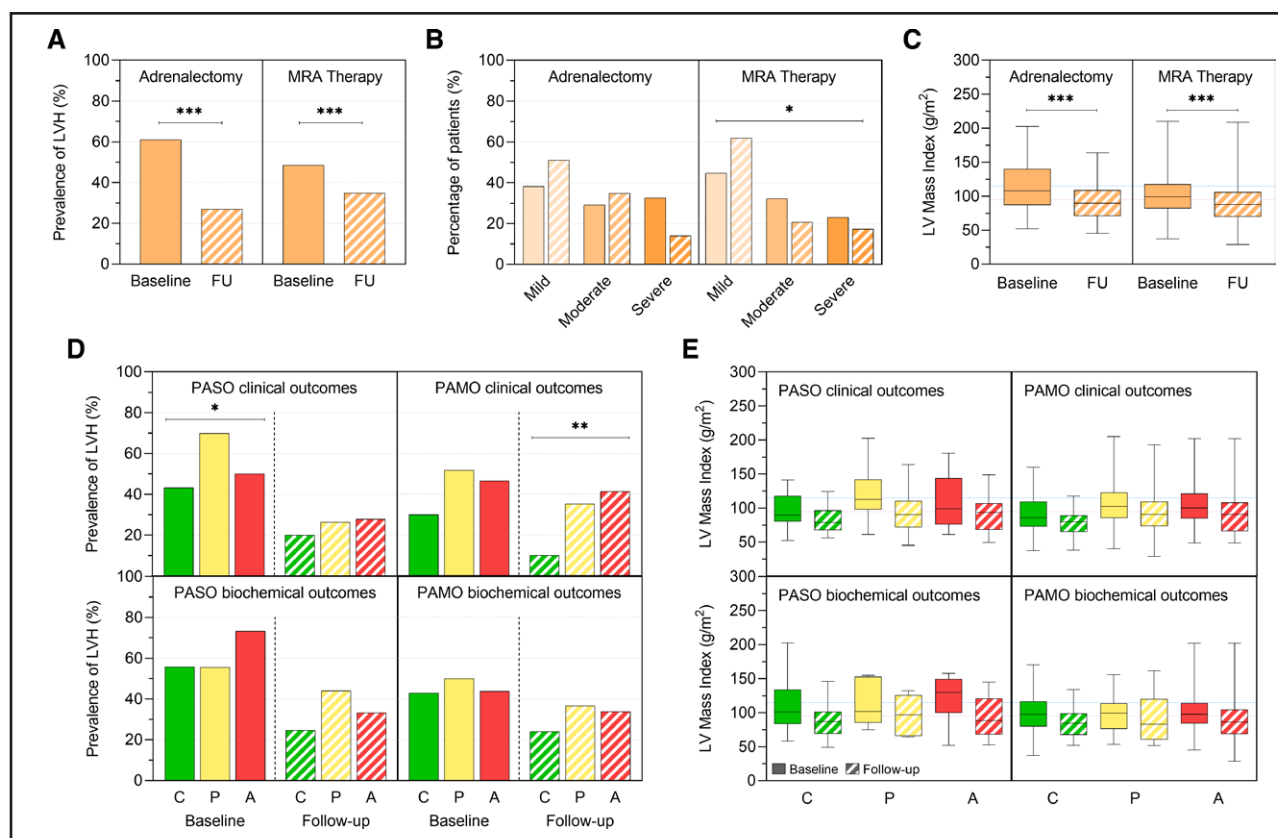
fully excluded. Of note, in the weighted linear regression, biochemical outcome was associated with LVMI variation, meaning a greater reduction in LVMI in patients with a complete biochemical response.

DISCUSSION

In the present study, we found that almost half of the patients with PA displayed LVH at baseline, and both systolic BP and PAC were independently associated with its identification and posttreatment regression. LVH regression at follow-up was observed in over half of the

patients, with a higher rate after adrenalectomy than after medical treatment, although the type of targeted treatment per se was not identified as a determinant of LVMI normalization. Complete PASO/PAMO biochemical success was associated with a trend toward greater LVMI reduction.

In line with previous reports,^{15,20,23,24} at baseline echocardiography, LVH was detected in $\approx 50\%$ of patients with PA. Current evidence suggests that the aldosterone excess observed in these patients contributes to cardiac remodeling through both hemodynamic mechanisms and direct hormonal effects on the myocardium.^{11–13,35}



Consistently, we found that both BP and PAC were independently associated with the presence of LVH, regardless of cardiometabolic comorbidities such as dyslipidemia, diabetes, or smoking habit, which have also been linked to cardiac hypertrophy.^{36–39} We also observed a higher prevalence of LVH in patients eligible for adrenalectomy; these subjects exhibited a more severe PA phenotype—higher BP, lower potassium, greater PAC, and a higher aldosterone-to-renin ratio—compared with those who underwent medical treatment. Thus, our findings support the contribution of both hypertension- and aldosterone-dependent mechanisms in the development of LVH^{14,18,21,24,29} and reinforce the concept that patients with untreated PA—especially those with a more severe clinical and biochemical phenotype^{14,15,21,23,24,40,41}—are at increased risk of adverse cardiac remodeling.

Obesity, diabetes, and dyslipidemia were associated with the presence of LVH in the overall baseline cohort of 1454 patients, and were similarly distributed between surgically and medically treated patients included in the longitudinal analysis, despite some prior studies reporting differences in the metabolic profile between the 2

treatment groups (or between unilateral and bilateral PA).^{42,43} Moreover, the prevalence of obesity and diabetes did not differ between patients with and without LVH regression after PA-specific treatment. These findings raise the possibility that the contribution of metabolic comorbidities to LVH regression may have been underestimated in our longitudinal analysis. Among the factors that may explain this finding, we should consider that patients with available follow-up echocardiography may not fully reflect the metabolic heterogeneity among patients with PA, and a small proportion of patients with unilateral disease were receiving medical treatment because of surgical contraindications or patient preference. In addition, metabolic characterization was limited to available clinical variables, whereas other relevant determinants (ie, insulin resistance, duration and treatment of diabetes, inflammatory markers, and changes in lifestyle) were not systematically available due to the retrospective design of the study.

Echocardiographic reassessment performed after at least 1 year of targeted treatment, with a median follow-up exceeding 3 years, revealed that both adrenalectomy

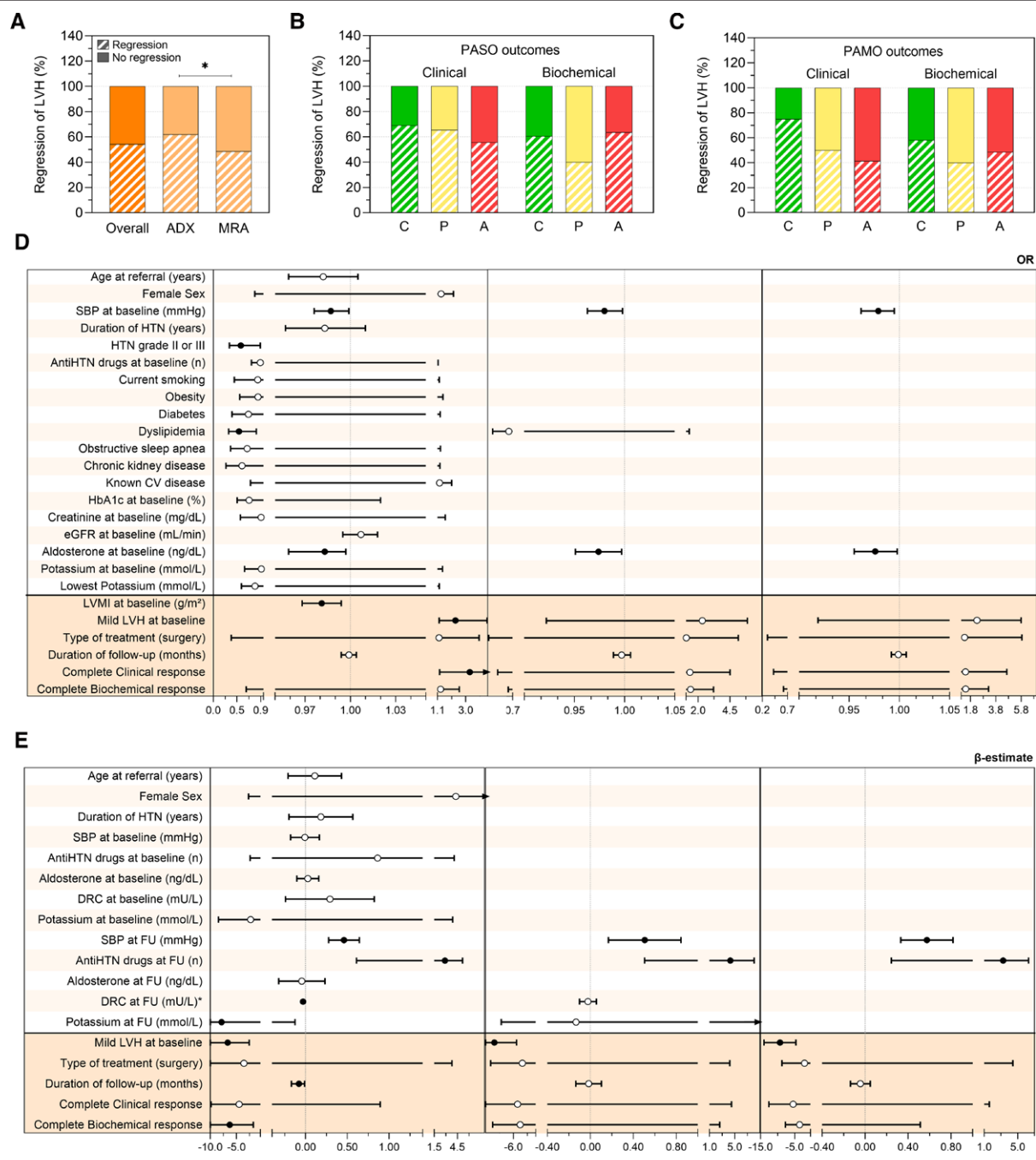


Figure 3. Regression of left ventricular hypertrophy (LVH) and clinical/biochemical determinants.

A, LVH regression in the overall cohort and after stratification by type of targeted treatment (adrenalectomy vs mineralocorticoid receptor antagonist [MRA]); * $P < 0.05$. **B**, LVH regression after stratification by clinical and biochemical outcomes of surgical treatment. **C**, LVH regression after stratification by clinical and biochemical responses to medical treatment. **D**, Determinants of LVH regression after at least 1 year of targeted treatment. Odds ratios (ORs) with 95% CIs are shown for univariable (first box from the left) and stepwise multivariable logistic regression analyses (second and third boxes represent Steps 1 and 2 of the adjusted multivariable model; see also Table S7). **E**, Factors associated with percentage variation in left ventricular mass index (LVMI) after at least 1 year of targeted treatment. β -estimates and 95% CIs are shown for univariable analysis and stepwise multivariable models (Steps 1 and 2, from left to right). For multivariable analyses assessing factors associated with LVH regression (**D**) and percentage changes in LVMI after targeted treatment (**E**), parameters associated with the outcome at univariable analysis were entered in Step 1, and those remaining associated with the outcome in Step 1 were entered in Step 2. In addition, a predefined set of clinically relevant variables, reported within the dark orange boxes, was included a priori as potential confounders. In the case of collinearity, only 1 variable was retained in the adjusted models. *Plasma renin activity (PRA) was converted to direct renin concentration (DRC) using a conversion factor ($K = 10$; $DRC = PRA \times K$). A indicates absent; ADX, adrenalectomy; C, complete; CV, cardiovascular; eGFR, estimated glomerular filtration rate; FU, follow-up; HbA1c, glycated hemoglobin A1c; HTN, hypertension; P, partial; PAMO, primary aldosteronism medical treatment outcomes; PASO, primary aldosteronism surgical outcomes; and SBP, systolic blood pressure.

Table 3. Baseline Characteristics of Patients Who Underwent LVH Regression

Variable	Regression (n=136)	Nonregression (n=115)	P value
Clinical parameters			
Age at referral, y	51±10.5	54±11.6	0.114
Female sex, n (%)	62 (45.6)	45 (39.1)	0.303
BMI, kg/m ²	28.2±5.48	28.9±4.52	0.324
BSA, m ²	1.99±0.276	2.00±0.225	0.801
SBP, mm Hg	153±21.0	159±23.4	0.028*
DBP, mm Hg	94±13.1	95±13.2	0.555
Duration of HTN, y	9 (4–15)	10 (2–17)	0.587
HTN grade II or III, n (%)	53 (46.1)	67 (62.0)	0.017*
Antihypertensive drugs, n	2 (1–3)	2 (2–3)	0.135
Hypokalemia at any time, n (%)	94 (69.1)	67 (58.3)	0.074
Potassium supplementation, n (%)	51 (37.5)	34 (30.4)	0.238
Current smoking, n (%)	16 (16.8)	18 (19.4)	0.654
Obesity, n (%)	45 (33.1)	42 (36.8)	0.535
Diabetes type 1 or 2, n (%)	23 (17.3)	27 (23.9)	0.200
Dyslipidemia, n (%)	45 (33.6)	58 (50.9)	0.006*
Obstructive sleep apnea, n (%)	19 (14.4)	23 (20.7)	0.194
Chronic kidney disease, n (%)	11 (8.1)	16 (13.9)	0.138
Known CV disease, n (%) [†]	48 (35.3)	36 (31.3)	0.504
Concurrent MACS, n (%)	7 (9.6)	10 (18.2)	0.156
Biochemical parameters			
Fasting plasma glucose, mg/dL	105±26.7	107±28.1	0.705
HbA1c, %	5.6±0.77	5.9±0.97	0.056
Creatinine, mg/dL	0.96±0.516	0.98±0.347	0.731
eGFR, mL/min	90±21.2	87±22.3	0.258
Total cholesterol, mg/dL	187±38.6	196±39.3	0.064
LDL cholesterol, mg/dL	111±33.8	118±34.3	0.091
HDL cholesterol, mg/dL	55±17.5	53±13.0	0.365
Triglycerides, mg/dL	118±71.4	127±61.0	0.307
Aldosterone, ng/dL	23.6 (14.3–35.9)	28.5 (20.5–41.8)	0.009*
PRA, ng/mL per h	0.30 (0.18–0.53)	0.20 (0.20–0.40)	0.343
DRC, mU/L	3.0 (2.0–7.1)	3.9 (2.0–10.1)	0.308
Potassium, mmol/L	3.6±0.53	3.6±0.64	0.679
Lowest potassium, mmol/L	3.3±0.62	3.4±0.67	0.248
Aldosterone after SIT, ng/dL	11.6 (7.5–22.9)	14.3 (9.0–25.3)	0.144
Aldosterone after CCT, ng/dL	19.1 (12.4–37.0)	19.6 (16.0–28.8)	0.650
Imaging (CT scan and MRI)			
Presence of nodules, n (%)	84 (62.2)	68 (62.4)	0.979
Largest size of nodules, mm	14 (10–19)	13 (10–19)	0.508
Adrenal venous sampling			
PA subtype at AVS, n (%)			
Unilateral	58 (52.8)	39 (47.0)	0.444
Bilateral	38 (34.5)	28 (33.7)	
Undetermined	14 (12.7)	16 (19.3)	
LI≥4 without or with ACTH	56 (56.6)	33 (50.8)	0.466

(Continued)

Table 3. Continued

Variable	Regression (n=136)	Nonregression (n=115)	P value
Echocardiographic parameters			
Severity of LVH, n (%)			
Mild	59 (48.0)	34 (35.1)	0.047*
Moderate	39 (31.7)	30 (30.9)	
Severe	25 (20.3)	33 (34.0)	
LV mass index, g/m ²	127±25.7	142±32.6	0.002*
IVS thickness, mm	13±2.4	13±2.2	0.119
PW thickness, mm	12±1.9	12±2.3	0.040*
LV ejection fraction, %	67±7.1	62±9.7	0.001*

Baseline characteristics of patients stratified according to LVH regression (defined as LVMI normalization) vs Nonregression (defined as persistent LVH) after at least 1 y of targeted treatment. ACTH indicates adrenocorticotropic hormone; ARR, aldosterone-to-renin ratio; AVS, adrenal venous sampling; BMI, body mass index; CCT, captopril-challenge test; CT, computed tomography; CV, cardiovascular; DBP, diastolic blood pressure; DRC, direct renin concentration; eGFR, estimated glomerular filtration rate; HbA1c, glycated hemoglobin A1c; HDL, high density lipoprotein; HTN, hypertension; IVS, interventricular septum; LDL, low density lipoprotein; LI, lateralization index; LV, left ventricular; LVH, left ventricular hypertrophy; LVMI, left ventricular mass index; MACS, mild autonomous cortisol secretion; MRI, magnetic resonance imaging; PA, primary aldosteronism; PRA, plasma renin activity; PW, posterior wall; SBP, systolic blood pressure; and SIT, saline infusion test.

*Excluding left ventricular hypertrophy.

and MRA therapy were associated with a significant improvement in LV structure. Overall, LVH regressed in 54.2% of patients, with a higher proportion of regression after adrenalectomy compared with medical therapy. However, the type of treatment was not associated with LVH regression or with the magnitude of LVMI reduction. Importantly, LVMI decreased after both surgical and medical treatment, with no significant differences in the percentage variation between groups, although a trend toward a greater reduction in LVMI was observed after adrenalectomy. The interventricular septal and LV posterior wall thickness did not show a significant reduction after medical therapy, although the magnitude of change was similar to that observed in surgically treated patients. The apparent discrepancy between LVMI and linear wall-thickness measurements in medically treated patients may reflect the greater sensitivity of LVMI to capture modest global reverse remodeling, whereas individual linear wall measurements may show smaller and more variable changes over time, particularly in a retrospective multicenter echocardiographic setting.

Taken together, these results indicate that both adrenalectomy and MRA therapy effectively reduce LVMI and can promote the reversal of cardiac remodeling in patients with PA at long-term follow-up, although surgical treatment may confer a greater likelihood of LVH regression. Of note, a higher MRA DDD emerged as an independent determinant of greater LVMI reduction; this finding is consistent with the concept that adequate mineralocorticoid receptor blockade may contribute to the reversal of aldosterone-mediated cardiac remodeling.⁴⁴ Comparative analyses of surgical *versus* medical treatment effect on LV mass have yielded conflicting results. While some studies report similar improvements

in LV mass following adrenalectomy or MRA therapy, others suggest a greater long-term reduction after surgery. A meta-analysis of 4 prospective studies including 355 patients with PA followed up for at least 6 months after targeted treatment found comparable reductions in LV mass between surgically- and medically treated patients.⁴⁵ More recently, 3 larger meta-analyses, examining prospective and retrospective PA cohorts, have suggested that adrenalectomy provides greater LV mass reduction than medical therapy, and could be more effective in achieving regression of both concentric and eccentric LVH.^{25–27} Notably, after stratification for the duration of follow-up, LVMI reduction with MRA therapy became similar to that achieved with adrenalectomy after >30 months.²⁷ Catena et al¹⁴ similarly suggested that adrenalectomy may produce earlier and greater regression of LVH compared with medical therapy in the first year of follow-up, although both strategies can lead to significant LVMI reduction over longer follow-up.

Beyond differences in study design, cohort characteristics, sample size, and follow-up duration, several factors may explain conflicting results on the effects of MRA versus surgical treatment on cardiac remodeling. The efficacy of MRA therapy may depend on disease duration and treatment intensity, as suggested by the association between MRA DDD, and LVMI reduction observed in our study. In patients treated with MRA, circulating aldosterone may remain elevated and contribute to adverse cardiac remodeling through its nongenomic effects.^{46,47} Aldosterone synthase inhibitors, which directly suppress aldosterone synthesis, may further improve cardiac remodeling beyond current medical treatment.⁴⁸

In our cohort, lower baseline BP and PAC were independently associated with LVH regression, regardless

of baseline LVH severity, treatment modality, clinical or biochemical success, and follow-up duration. This indicates that patients with a more severe PA phenotype at diagnosis are less likely to achieve full reversal of cardiac damage. Following the concept of a continuum spectrum of PA,⁴⁹ ranging from mild renin-independent aldosterone excess to overt disease, early diagnosis and timely treatment may help mitigate both hemodynamic and aldosterone-related cardiac remodeling. Conversely, the magnitude of LVMI reduction was primarily associated with a lower systolic BP at follow-up, fewer antihypertensive medications required, and a milder baseline LVH severity, again independently of treatment modality, clinical or biochemical outcomes, and follow-up duration. Therefore, patients with more advanced remodeling or suboptimal BP control after targeted treatment may undergo more modest improvements in LVMI.

Finally, we evaluated the impact of the application of standardized PASO and PAMO outcome definitions on pre- and posttreatment changes in LV structure. Complete PASO clinical responders were less likely to display LVH at baseline than partial or absent responders, with a similar trend across PAMO clinical outcome categories. These findings are consistent with the results of PASO and PAMO cohort studies,^{17,30} which identified the presence of LVH at baseline as a predictor of complete clinical success after adrenalectomy or MRA therapy. In our study, follow-up echocardiographic parameters did not differ after stratification for PASO and PAMO outcomes, although there was a trend toward a greater improvement in cardiac remodeling among complete and partial responders. However, the univariable unadjusted analyses suggested that achieving a complete clinical response was associated with a higher likelihood of LVH regression, whereas a successful biochemical response, together with higher follow-up potassium or renin levels, was associated with greater LVMI reduction. The lack of an independent association between PASO/PAMO outcomes and LV structural changes may depend on additional or unmeasured factors, such as MRA adherence, dietary salt intake, or physical activity, which could have influenced both echocardiographic and clinical/biochemical parameters.^{25,50}

Of note, adequate BP control has been shown to promote regression of LVH in PA,²⁰ but evidence linking posttreatment biochemical features to cardiac remodeling remains limited. Previous studies have reported higher posttreatment renin levels in patients who achieved LVMI normalization following surgical or medical treatment,²¹ and correlations between postsurgical LVMI variation and postoperative increases in serum potassium levels and renin activity following adrenalectomy.⁵¹ Similarly, other authors reported higher rates of LVH regression and greater reductions in interventricular septum and LV posterior wall thickness in patients with unsuppressed renin concentrations during MRA therapy as compared

with those with persistently suppressed renin²³; additionally, in patients with unsuppressed posttreatment renin, follow-up potassium and renin levels independently predicted greater LVMI reductions.²³ In accordance with the results of these studies, our data support the importance of achieving a full biochemical response to improve long-term cardiac remodeling in patients with PA.

This study has some limitations that should be acknowledged. First, its retrospective design did not allow us to infer a causal relationship between individual predictors and LVH regression. Data on classes of antihypertensive medications were not available in all patients and therefore could not be included in the analyses. Finally, the impact of BP may have been underestimated because office BP measurements were used rather than 24-hour ambulatory BP monitoring, which better predicts LV mass.⁵² Nevertheless, we recruited a multicenter international cohort that represents the largest population of patients with PA analyzed to identify determinants of LVH before and after targeted treatment, thereby improving the generalizability of previous single-center observations. The prolonged median follow-up period enabled a robust evaluation of changes in LVMI over time, with a comparative impact of adrenalectomy *versus* MRA therapy. Furthermore, we systematically applied PASO and PAMO criteria for the first time, ensuring a consistent and reproducible classification of treatment responses, and facilitating the comparisons of cardiac remodeling between treatment modalities and across different outcome categories. Importantly, our data support the concept that prolonged MRA therapy may also promote LVH regression, rather than simple stabilization of LV mass. Finally, the association between higher MRA DDD and greater LVMI reduction in medically treated patients highlights the importance of adequate mineralocorticoid receptor blockade in reversing aldosterone-mediated cardiac damage.

In conclusion, our study confirms the high prevalence of LVH in patients with PA and shows that higher baseline aldosterone concentrations are independently associated both with the presence of LVH and with a lower likelihood of LVH regression after targeted treatment, independently of BP. Our findings further support the observation that both adrenalectomy and long-term MRA therapy can induce LVH regression.

PERSPECTIVES

The present study underscores the importance of a comprehensive echocardiographic assessment both at baseline and follow-up in patients with PA, to improve their cardiovascular risk stratification. Alongside baseline PA phenotype, the definition of clinical and biochemical outcomes according to PASO and PAMO criteria may help clinicians identify patients who are more likely to achieve LVH regression. However, further research is needed to

clarify their relationship with posttreatment changes in LV structure.

ARTICLE INFORMATION

Received February 15, 2026; accepted July 28, 2026.

Affiliations

Department of Endocrinology and Nutrition (J.G., M.A.-C.) and Department of Biochemistry (A.M.G.C., L.C.G.), Hospital Universitario Ramón y Cajal, Madrid, Spain. Division of Internal Medicine and Hypertension Unit, Department of Medical Sciences, University of Torino, Italy (J.B., V.C., P.M.). Division of Endocrinology, Department of Medicine IV (H.S., E.J.P., S.Z., M. Reincke) and Division of Vascular Medicine, Department of Medicine IV (H.S.), LMU University Hospital, Munich, Germany. Department of Endocrinology and Nutrition, Hospital Universitario Son Espases, Palma, Islas Baleares, Spain (E.M.R., A.S.S.). Department of Endocrinology and Nutrition, Hospital Universitario La Paz, Madrid, Spain (P.P.R., P.M.R.-M.). Department of Endocrinology and Nutrition, OSI Bilbao-Basurto, Hospital Universitario de Basurto, Spain (M.P.F.). Department of Endocrinology and Nutrition, Hospital Universitario Infanta Sofía, Madrid, Spain (P.D.G.). Department of Endocrinology and Nutrition, Hospital Universitario Reina Sofía, Córdoba, Spain (A.R.R.). Department of Internal Medicine, Hospital Universitario Infanta Leonor, Madrid, Spain (E.M.M.M.). Department of Endocrinology and Nutrition, Hospital Universitario Fundación Jiménez Díaz, Madrid, Spain (J.G.R.-S.). Department of Endocrinology and Nutrition, Hospital Universitario de Castellón, Spain (M.G.-B.). Department of Internal Medicine, Hospital Universitario Virgen de las Nieves, Granada, Spain (F.J.A.). Department of Endocrinology and Nutrition, Clínica Universidad de Navarra, Pamplona, Spain (C.M.P.). Department of Endocrinology and Nutrition, Hospital Clínico Universitario de Salamanca, Spain (C.R.L.). Department of Endocrinology and Nutrition, Hospital de Girona Doctor Josep Trueta, Spain (M. Recasens, R.B.S.M.). Department of Endocrinology and Nutrition, Hospital Universitario de Valladolid, Spain (E.G.-H.). Department of Endocrinology and Nutrition, Hospital Universitario Virgen de la Victoria de Málaga, IBIMA, Spain (M.J.P.C.). Department of Endocrinology and Nutrition, Clínica Universidad de Navarra, Madrid, Spain (S.G.-M.). Department of Endocrinology and Nutrition, Hospital Universitario de Albacete, Spain (C.L.). Department of Endocrinology and Nutrition, Hospital Universitario Val d'Hebron, Barcelona, Spain (A.C.). Instituto de Investigación Biomédica Ramón y Cajal (IRYCIS), Madrid, Spain (M.A.-C.). University of the Basque Country UPV/EHU, Bilbao, Spain (M.P.F.).

Sources of Funding

This study was funded by the Spanish Society of Endocrinology & Nutrition (SEEN).

Disclosures

None.

REFERENCES

- Monticone S, Burrello J, Tizzani D, Bertello C, Viola A, Buffolo F, Gabetti L, Mengozzi G, Williams TA, Rabbia F, et al. Prevalence and clinical manifestations of primary aldosteronism encountered in primary care practice. *J Am Coll Cardiol*. 2017;69:1811–1820. doi: 10.1016/j.jacc.2017.01.052
- Xu Z, Yang J, Hu J, Song Y, He W, Luo T, Cheng Q, Ma L, Luo R, Fuller PJ, et al; Chongqing Primary Aldosteronism Study (CONPASS) Group. Primary aldosteronism in patients in China with recently detected hypertension. *J Am Coll Cardiol*. 2020;75:1913–1922. doi: 10.1016/j.jacc.2020.02.052
- Calhoun DA, Nishizaka MK, Zaman MA, Thakkar RB, Weissmann P. Hyperaldosteronism among black and white subjects with resistant hypertension. *Hypertension*. 2002;40:892–896. doi: 10.1161/01.hyp.0000040261.30455.b6
- Rossi GP, Bernini G, Caliumi C, Desideri G, Fabris B, Ferri C, Ganzaroli C, Giacchetti G, Letizia C, Maccario M, et al; PAPY Study Investigators. A prospective study of the prevalence of primary aldosteronism in 1,125 hypertensive patients. *J Am Coll Cardiol*. 2006;48:2293–2300. doi: 10.1016/j.jacc.2006.07.059
- Reincke M, Fischer E, Gerum S, Merkle K, Schulz S, Pallauf A, Quinkler M, Hanslik G, Lang K, Hahner S, et al; German Conn's Registry–Else Kröner-Fresenius-Hyperaldosteronism Registry. Observational study mortality in treated primary aldosteronism: the German Conn's registry. *Hypertension*. 2012;60:618–624. doi: 10.1161/HYPERTENSIONAHA.112.197111
- Monticone S, D'Ascenzo F, Moretti C, Williams TA, Veglio F, Gaita F, Mulatero P. Cardiovascular events and target organ damage in primary aldosteronism compared with essential hypertension: a systematic review and meta-analysis. *Lancet Diabetes Endocrinol*. 2018;6:41–50. doi: 10.1016/S2213-8587(17)30319-4
- Inoue K, Naito T, Fuji R, Sonehara K, Yamamoto K, Baba R, Kodama T, Otagaki Y, Okada A, Itcho K, et al; BioBank Japan. Primary aldosteronism and risk of cardiovascular outcomes: genome-wide association and Mendelian randomization study. *J Am Heart Assoc*. 2024;13:e034180. doi: 10.1161/JAHA.123.034180
- Schillaci G, Verdecchia P, Porcellati C, Cuccurullo O, Cosco C, Perticone F. Continuous relation between left ventricular mass and cardiovascular risk in essential hypertension. *Hypertension*. 2000;35:580–586. doi: 10.1161/01.hyp.35.2.580
- Bombelli M, Facchetti R, Carugo S, Madotto F, Arenare F, Quarti-Trevano F, Capra A, Giannattasio C, Dell'Oro R, Grassi G, et al. Left ventricular hypertrophy increases cardiovascular risk independently of in-office and out-of-office blood pressure values. *J Hypertens*. 2009;27:2458–2464. doi: 10.1097/HJH.0b013e328330b845
- Bombelli M, Vanoli J, Facchetti R, Maloberti A, Cuspidi C, Grassi G, Mancia G. Impact of the increase in left ventricular mass on the risk of long-term cardiovascular mortality: a prospective cohort study. *Hypertension*. 2023;80:1321–1330. doi: 10.1161/HYPERTENSIONAHA.122.19988
- Rossi GP, Di Bello V, Ganzaroli C, Sacchetto A, Cesari M, Bertini A, Giorgi D, Scognamiglio R, Mariani M, Pessina AC. Excess aldosterone is associated with alterations of myocardial texture in primary aldosteronism. *Hypertension*. 2002;40:23–27. doi: 10.1161/01.hyp.0000023182.68420.eb
- Inoue K, Goldwater D, Allison M, Seeman T, Kestenbaum BR, Watson KE. Serum aldosterone concentration, blood pressure, and coronary artery calcium: the multi-ethnic study of atherosclerosis. *Hypertension*. 2020;76:113–120. doi: 10.1161/HYPERTENSIONAHA.120.15006
- Ohno Y, Sone M, Inagaki N, Yamasaki T, Ogawa O, Takeda Y, Kurihara I, Itoh H, Umakoshi H, Tsuiji M, et al; Nagahama Study. Prevalence of cardiovascular disease and its risk factors in primary aldosteronism: a multicenter study in Japan. *Hypertension*. 2018;71:530–537. doi: 10.1161/HYPERTENSIONAHA.117.10263
- Catena C, Colussi G, Lapenna R, Nadalin E, Chiuch A, Gianfagna P, Sechi LA. Long-term cardiac effects of adrenalectomy or mineralocorticoid antagonists in patients with primary aldosteronism. *Hypertension*. 2007;50:911–918. doi: 10.1161/HYPERTENSIONAHA.107.095448
- Muesan ML, Salvetti M, Paini A, Agabiti-Rosei C, Monteduro C, Galbassini G, Belotti E, Aggusti C, Rizzoni D, Castellano M, et al. Inappropriate left ventricular mass in patients with primary aldosteronism. *Hypertension*. 2008;52:529–534. doi: 10.1161/HYPERTENSIONAHA.108.114140
- Adler GK, Stowasser M, Correa RR, Khan N, Kline G, McGowan MJ, Mulatero P, Murad MH, Touyz RM, Vaidya A, et al. Primary Aldosteronism: an Endocrine Society Clinical Practice Guideline. *J Clin Endocrinol Metab*. 2025;110:2453–2495. doi: 10.1210/clinem/dgaf284
- Williams TA, Lenders JWM, Mulatero P, Burrello J, Rottenkolber M, Adolf C, Satoh F, Amar L, Quinkler M, Deinum J, et al; Primary Aldosteronism Surgery Outcome (PASO) investigators. Outcomes after adrenalectomy for unilateral primary aldosteronism: an international consensus on outcome measures and analysis of remission rates in an international cohort. *Lancet Diabetes Endocrinol*. 2017;5:689–699. doi: 10.1016/S2213-8587(17)30135-3
- Giacchetti G, Ronconi V, Turchi F, Agostinelli L, Mantero F, Rilli S, Boscaro M. Aldosterone as a key mediator of the cardiometabolic syndrome in primary aldosteronism: an observational study. *J Hypertens*. 2007;25:177–186. doi: 10.1097/HJH.0b013e3280108e6f
- Bernini G, Bacca A, Carli V, Carrara D, Materazzi G, Berti P, Miccoli P, Pisano R, Tantarini V, Bernini M, et al. Cardiovascular changes in patients with primary aldosteronism after surgical or medical treatment. *J Endocrinol Invest*. 2012;35:274–280. doi: 10.3275/7611
- Rossi GP, Cesari M, Cuspidi C, Maiolino G, Cicala MV, Bisogni V, Mantero F, Pessina AC. Long-term control of arterial hypertension and regression of left ventricular hypertrophy with treatment of primary aldosteronism. *Hypertension*. 2013;62:62–69. doi: 10.1161/HYPERTENSIONAHA.113.01316
- Ueda T, Tsurutani Y, Osada J, Inoue K, Hoshino Y, Ono M, Nakai K, Saito J, Yumoto K, Nishikawa T. Comparison of echocardiographic changes between surgery and medication treatment in patients with primary aldosteronism. *J Am Heart Assoc*. 2022;11:e023813. doi: 10.1161/JAHA.121.023813
- Cuspidi C, Tadic M, Sala C, Quarti-Trevano F, Gherbesi E, Mancia G, Grassi G. Regression of left ventricular hypertrophy in primary aldosteronism after adrenalectomy: a meta-analysis of echocardiographic studies. *J Hypertens*. 2021;39:775–783. doi: 10.1097/HJH.0000000000002679
- Köhler A, Sarkis AL, Heinrich DA, Müller L, Handgriff L, Deniz S, Schneider H, Künzel H, Ladurner R, Reincke M, et al. Renin, a marker for left ventricular

- hypertrophy, in primary aldosteronism: a cohort study. *Eur J Endocrinol*. 2021;185:663–672. doi: 10.1530/EJE-21-0018
24. Ohno Y, Sone M, Inagaki N, Kawashima A, Takeda Y, Yoneda T, Kurihara I, Itoh H, Tsuike M, Ichijo T, et al; JPAS/JRAS Study Group. Nadir aldosterone levels after confirmatory tests are correlated with left ventricular hypertrophy in primary aldosteronism. *Hypertension*. 2020;75:1475–1482. doi: 10.1161/HYPERTENSIONAHA.119.14601
 25. Marzano L, Zoccatelli F, Pizzolo F, Friso S. Adrenalectomy versus medical therapy in primary aldosteronism: a meta-analysis of long-term cardiac remodeling and function: medical versus adrenalectomy treatment compared in hyperaldosteronism (MATCH) study. *Hypertension*. 2025;82:e142–e153. doi: 10.1161/HYPERTENSIONAHA.125.25104
 26. Cheng J, Ma R, Dai R, Ran J, Wang B. Comparison of cardiac structural improvement in patients with primary aldosteronism after surgical therapy and drug therapy: a meta-analysis. *QJEMD*. 2022;12:283–295. doi: 10.4236/ojemd.2022.1212021
 27. Chang YC, Chen UL, Hung CS, Chen ZW, Liao CW, Brown JM, Vaidya A, Wu VC, Tsai CH, Lin YH. Evaluating the impact of adrenalectomy and mineralocorticoid receptor antagonists for primary aldosteronism on left ventricular remodeling: a systematic review and meta-analysis. *Eur J Endocrinol*. 2026;194:S1–S13. doi: 10.1093/ejendo/lvaf271
 28. Rossi GP, Sacchetto A, Visentin P, Canali C, Graniero GR, Palatini P, Pessina AC. Changes in left ventricular anatomy and function in hypertension and primary aldosteronism. *Hypertension*. 1996;27:1039–1045. doi: 10.1161/01.hyp.27.5.1039
 29. Catena C, Colussi GL, Marzano L, Sechi LA. Predictive factors of left ventricular mass changes after treatment of primary aldosteronism. *Horm Metab Res*. 2012;44:188–193. doi: 10.1055/s-0032-1301902
 30. Yang J, Burrello J, Goi J, Reincke M, Adolf C, Asbach E, Brüdgam D, Li Q, Song Y, Hu J, et al. Outcomes after medical treatment for primary aldosteronism: an international consensus and analysis of treatment response in an international cohort. *Lancet Diabetes Endocrinol*. 2025;13:119–133. doi: 10.1016/S2213-8587(24)00308-5
 31. Goi J, Paja Fano M, Rizo Gellida A, González-Boillos M, Rojas-Marcos PM, Caja Guayervas L, García Cano AM, Ruiz-Sánchez JG, Vicente A, Gómez-Hoyos E, et al. Efficacy and safety of medical treatment of primary aldosteronism: a real-world data study in Spain. *Eur J Endocrinol*. 2025;193:297–309. doi: 10.1093/ejendo/lvaf158
 32. Funder JW, Carey RM, Mantero F, Murad MH, Reincke M, Shibata H, Stowasser M, Young WF Jr. The management of primary aldosteronism: case detection, diagnosis, and treatment: an endocrine society clinical practice guideline. *J Clin Endocrinol Metab*. 2016;101:1889–1916. doi: 10.1210/jc.2015-4061
 33. Mulatero P, Monticone S, Deinum J, Amar L, Prejbisz A, Zennaro MC, Beuschlein F, Rossi GP, Nishikawa T, Morganti A, et al. Genetics, prevalence, screening and confirmation of primary aldosteronism: a position statement and consensus of the Working Group on Endocrine Hypertension of The European Society of Hypertension. *J Hypertens*. 2020;38:1919–1928. doi: 10.1097/HJH.0000000000002510
 34. Lang RM, Badano LP, Mor-Avi V, Afkalo J, Armstrong A, Ernande L, Flachskampf FA, Foster E, Goldstein SA, Kuznetsova T, et al. Recommendations for cardiac chamber quantification by echocardiography in adults: an update from the American Society of Echocardiography and the European Association of Cardiovascular Imaging. *J Am Soc Echocardiogr*. 2015;28:1–39.e14. doi: 10.1016/j.jecho.2014.10.003
 35. Burrello J, Senesi G, Altomare C, Buffolo F, D'Ascenzo F, Barile L, Mulatero P, Balbi C, Monticone S. Human 3D cardiac microtissue model to investigate aldosterone-induced fibrosis and electrical dysfunction. *Hypertension*. 2026;83:e25686. doi: 10.1161/HYPERTENSIONAHA.125.25686
 36. Horio T, Miyazato J, Kamide K, Takiuchi S, Kawano Y. Influence of low high-density lipoprotein cholesterol on left ventricular hypertrophy and diastolic function in essential hypertension. *Am J Hypertens*. 2003;16(11 Pt 1):938–944. doi: 10.1016/s0895-7061(03)01015-x
 37. Chen UL, Liao CW, Wang SM, Lai TS, Huang KH, Chang CC, Lee BC, Lu CC, Chang YR, Chang YY, et al; TAIPAI study group. Diabetes mellitus is associated with more adverse non-hemodynamic left ventricular remodeling and less recovery in patients with primary aldosteronism. *J Invest Med*. 2023;71:101–112. doi: 10.1177/10815589221141840
 38. Eguchi K, Boden-Albala B, Jin Z, Rundek T, Sacco RL, Homma S, Di Tullio MR. Association between diabetes mellitus and left ventricular hypertrophy in a multiethnic population. *Am J Cardiol*. 2008;101:1787–1791. doi: 10.1016/j.amjcard.2008.02.082
 39. Al Hariri M, Zibara K, Farhat W, Hashem Y, Soudani N, Al Ibrahim F, Hamade E, Zeidan A, Husari A, Kobeissy F. Cigarette smoking-induced cardiac hypertrophy, vascular inflammation and injury are attenuated by antioxidant supplementation in an animal model. *Front Pharmacol*. 2016;7:397. doi: 10.3389/fphar.2016.00397
 40. Lin YH, Wang SM, Wu VC, Lee JK, Kuo CC, Yen RF, Liu KL, Huang KH, Chueh SC, Wang WJ, et al; TAIPAI study group. The association of serum potassium level with left ventricular mass in patients with primary aldosteronism. *Eur J Clin Invest*. 2011;41:743–750. doi: 10.1111/j.1365-2362.2010.02462.x
 41. Hung CS, Ho YL, Chang YY, Wu VC, Wu XM, Lee JK, Chueh SC, Lin YH, Chang YS, Yang SY, et al. Twenty-four-hour urinary aldosterone predicts inappropriate left ventricular mass index in patients with primary aldosteronism. *ScientificWorldJournal*. 2013;2013:294594. doi: 10.1155/2013/294594
 42. Ohno Y, Sone M, Inagaki N, Yamasaki T, Ogawa O, Takeda Y, Kurihara I, Umakoshi H, Ichijo T, Katabami T, et al; JPAS Study Group. Obesity as a key factor underlying idiopathic hyperaldosteronism. *J Clin Endocrinol Metab*. 2018;103:4456–4464. doi: 10.1210/jc.2018-00866
 43. Spyroglou A, Handgriff L, Müller L, Schwarzmüller P, Parasiliti-Caprino M, Fuss CT, Remde H, Hirsch A, O'Toole SM, Thuzar M, et al. The metabolic phenotype of patients with primary aldosteronism: impact of subtype and sex - a multicenter-study of 3566 Caucasian and Asian subjects. *Eur J Endocrinol*. 2022;187:361–372. doi: 10.1530/EJE-22-0040
 44. Alfarano M, Marchionni G, Costantino J, Ballatore F, Verardo R, Miraldi F, Ciciarello FL, Petramala L, Letizia C, Frustaci A, et al. Aldosterone-related cardiovascular disease and benefits of mineralocorticoid receptor antagonists in clinical practice. *JACC Adv*. 2025;4:101762. doi: 10.1016/j.jaccadv.2025.101762
 45. Marzano L, Colussi G, Sechi LA, Catena C. Adrenalectomy is comparable with medical treatment for reduction of left ventricular mass in primary aldosteronism: meta-analysis of long-term studies. *Am J Hypertens*. 2015;28:312–318. doi: 10.1093/ajh/hpu154
 46. Brown NJ. Contribution of aldosterone to cardiovascular and renal inflammation and fibrosis. *Nat Rev Nephrol*. 2013;9:459–469. doi: 10.1038/nrneph.2013.110
 47. Brown JM. Adverse effects of aldosterone: beyond blood pressure. *J Am Heart Assoc*. 2024;13:e030142. doi: 10.1161/JAHA.123.030142
 48. Turcu AF, Freeman MW, Bancos I, Ben-Shlomo A, Hamidi O, Hamrahian AH, Huang W, Kirschner LS, Sam R, Mallappa A, et al; SPARK Investigator Group. Phase 2a study of baxdrostat in primary aldosteronism. *N Engl J Med*. 2025;393:515–518. doi: 10.1056/NEJMc2508629
 49. Vaidya A, Mulatero P, Baudrand R, Adler GK. The expanding spectrum of primary aldosteronism: implications for diagnosis, pathogenesis, and treatment. *Endocr Rev*. 2018;39:1057–1088. doi: 10.1210/er.2018-00139
 50. Bu X, Sun F, Li Q, Zhang H, Jiang N, Luo Y, Zhao Z, Yan Z, He H, Zhu Z. High salt intake exacerbates aldosterone-related target organ damage in patients with primary aldosteronism. *Hypertens Res*. 2025;48:2339–2348. doi: 10.1038/s41440-025-02270-4
 51. Liao CW, Chen A, Lin YT, Chang YY, Wang SM, Wu VC, Hung CS, Wu KD, Chueh SC, Lin YH; TAIPAI Study Group. The relation between the degree of left ventricular mass regression and serum potassium level change in patients with primary aldosteronism after adrenalectomy. *J Invest Med*. 2015;63:816–820. doi: 10.1097/JIM.0000000000000215
 52. Mancia G, Zanchetti A, Agabiti-Rosei E, Benemio G, De Cesaris R, Fogari R, Pessina A, Porcellati C, Rappelli A, Salvetti A, et al. Ambulatory blood pressure is superior to clinic blood pressure in predicting treatment-induced regression of left ventricular hypertrophy. SAMPLE Study Group. Study on Ambulatory Monitoring of Blood Pressure and Lisinopril Evaluation. *Circulation*. 1997;95:1464–1470. doi: 10.1161/01.cir.95.6.1464
 53. Mancia G, Kreutz R, Brunström M, Burnier M, Grassi G, Januszewicz A, Muiesan ML, Tsioufis K, Agabiti-Rosei E, Algharably EAE, et al. 2023 ESH guidelines for the management of arterial hypertension the task force for the management of arterial hypertension of the European society of hypertension: endorsed by the International Society of Hypertension (ISH) and the European Renal Association (ERA). *J Hypertens*. 2023;41:1874–2071. doi: 10.1097/HJH.00000000000003480
 54. Fassnacht M, Tsarakis S, Terzolo M, Tabarin A, Sahdev A, Newell-Price J, Pelsma I, Marina L, Lorenz K, Bancos I, et al. European Society of Endocrinology clinical practice guidelines on the management of adrenal incidentalomas, in collaboration with the European Network for the Study of Adrenal Tumors. *Eur J Endocrinol*. 2023;189:G1–G42. doi: 10.1093/ejendo/lvad066